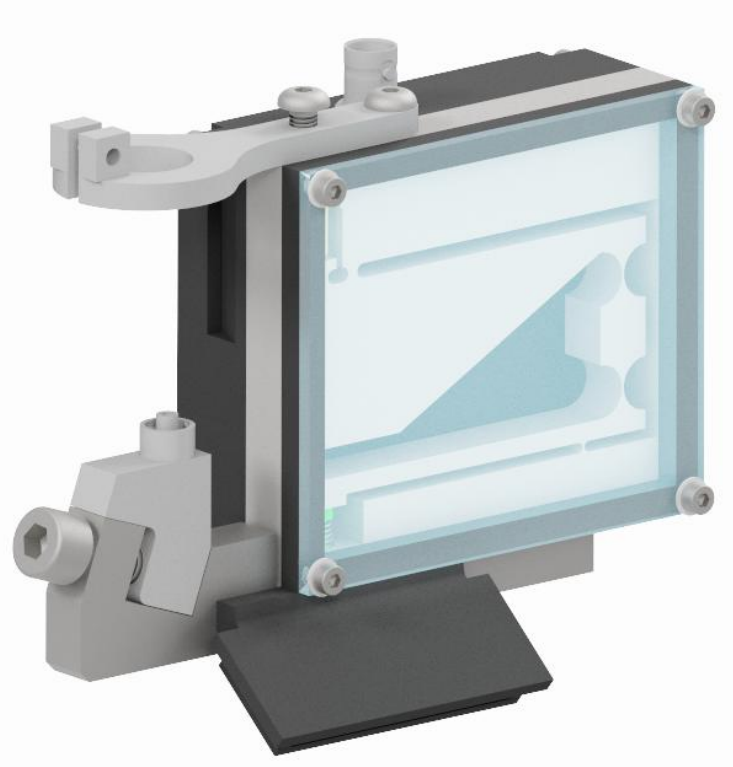




R2R – Spotter Plotter

Bachelor Project

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Co-supervisor:
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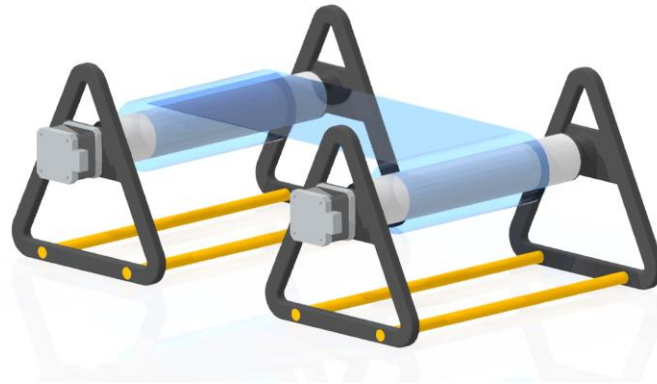
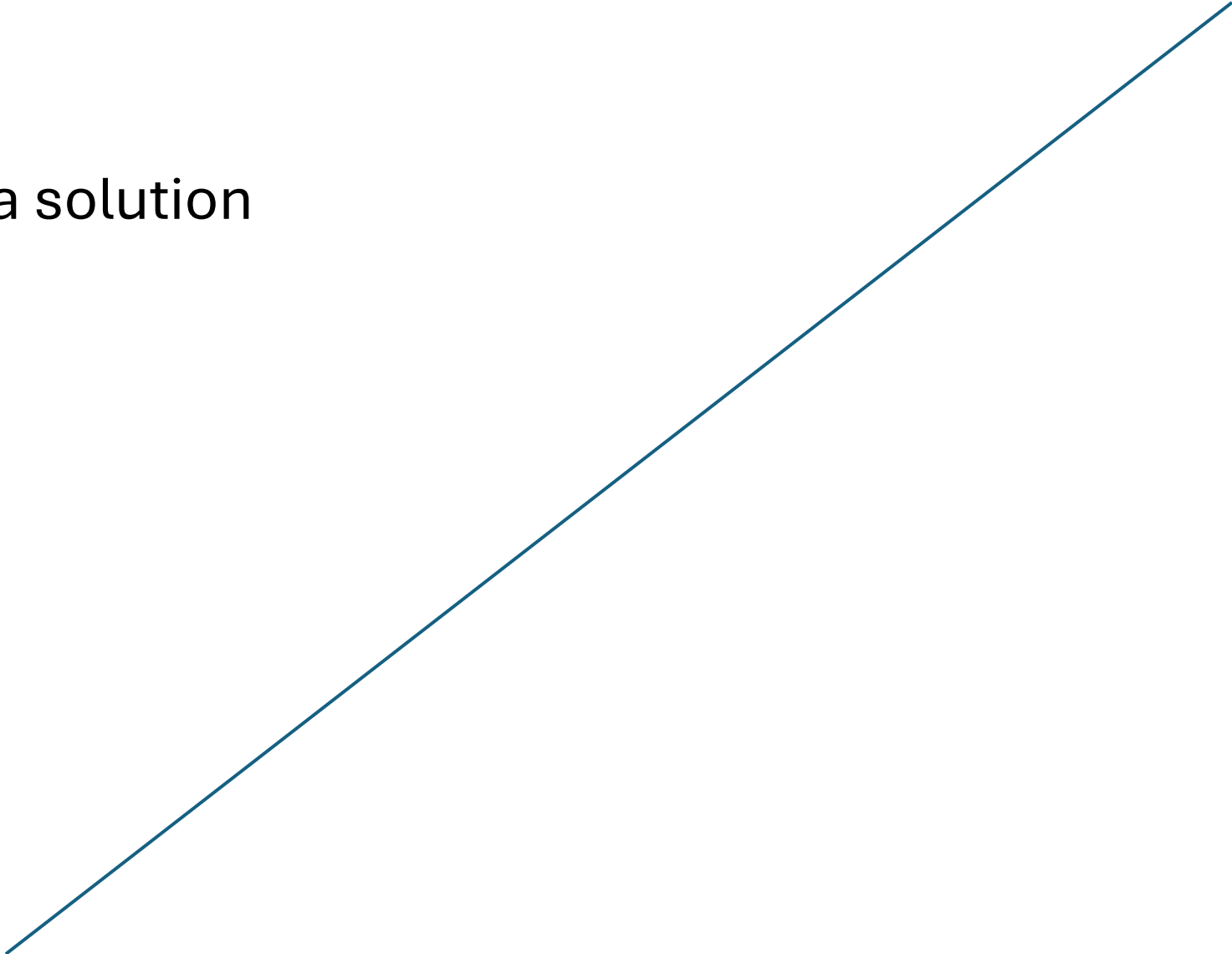


Table of contents:

- Problem description
 - Jet-Valve – R2R system as a solution
 - The add-on requirement
 - Dry run presentation
 - Problems of the system
 - How to improve
- 

Problem description:

Creation of optodes:

- “Deposition of multiple chemicals in various pattern”
- Aim for highest resolution achievable

Problem description:

Contact Spotting offers:

- Absolute positioning
- High viscosity support
- No affection by directional hysteresis
- High repeatability

Achieved:

- 300 micron diameter
- 600 micron pitch



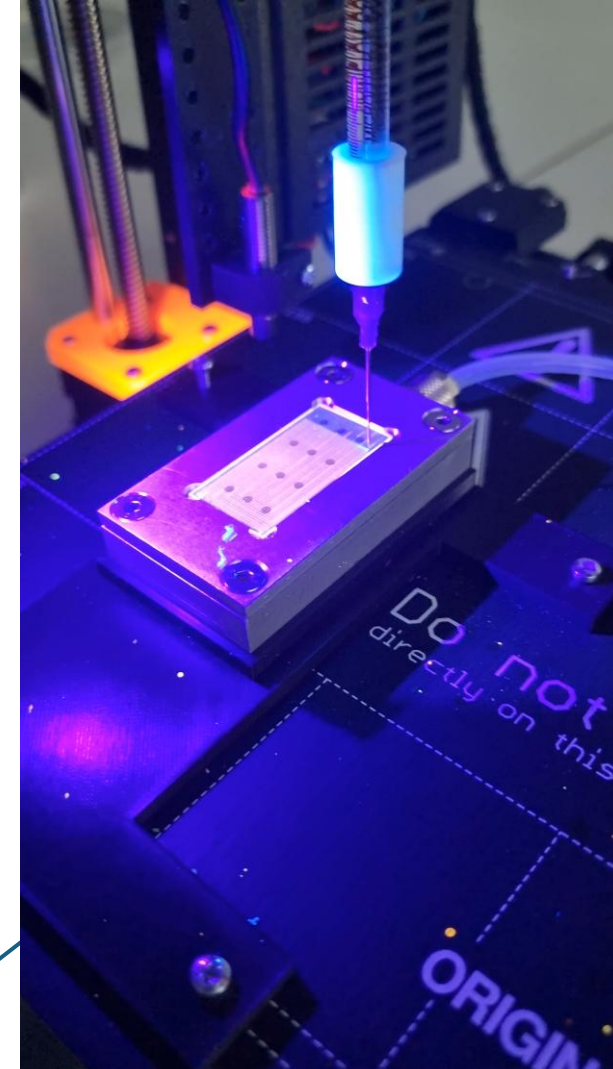
Problem description:

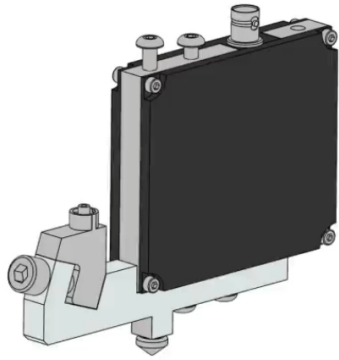
Contact Spotting requires:

- Precise height control (100-200 micron)
- Flat surface ± 37.5 microns
- Repetitive up and down motion

And is problematic for:

- **Scalability** (slow because of repetitive motion)
- **Safety** (miscalibration risks needle crashing)





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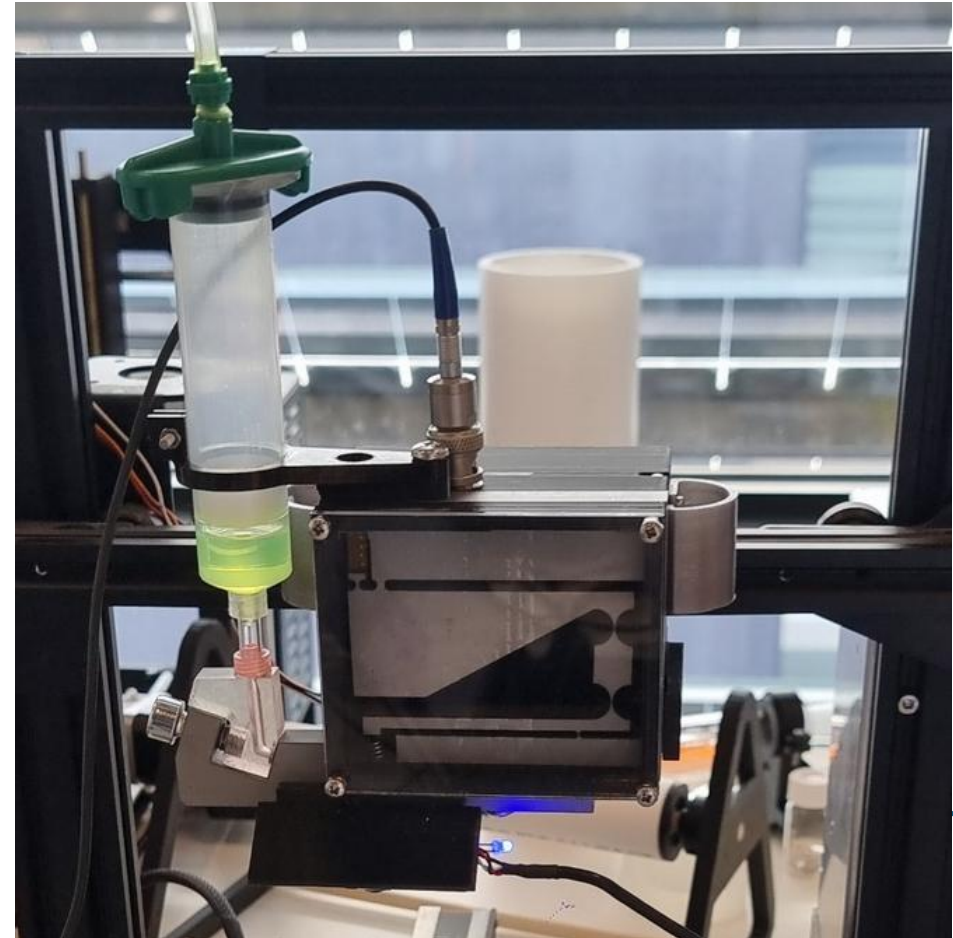
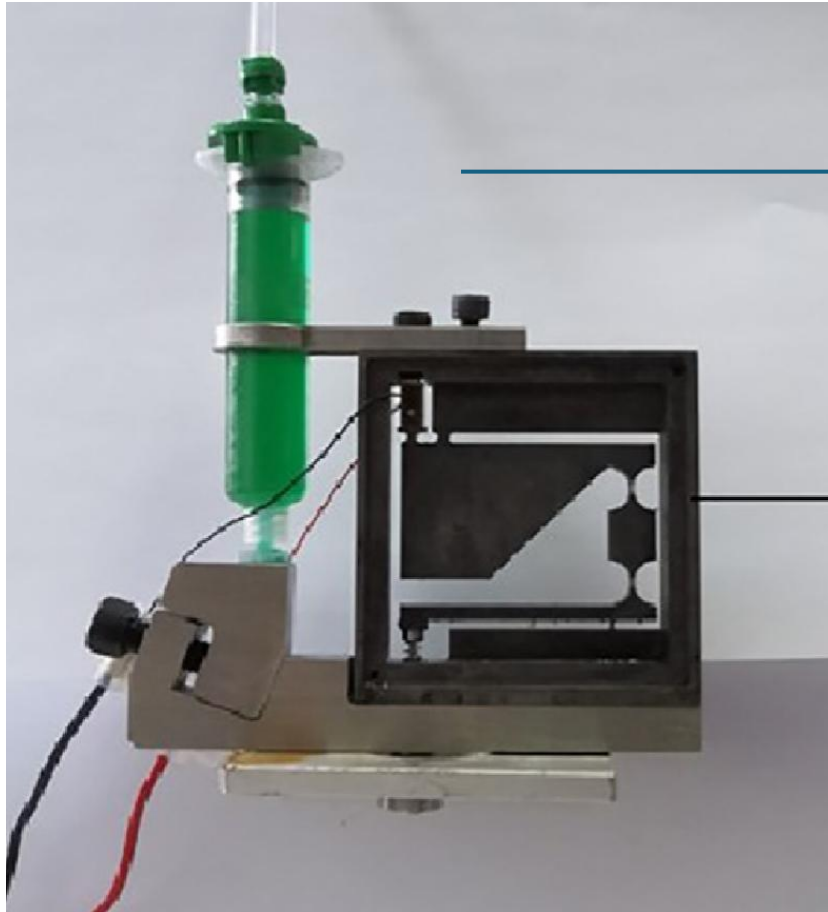
R2R-Spotter Plotter:

combining a Jet-Valve with a R2R system

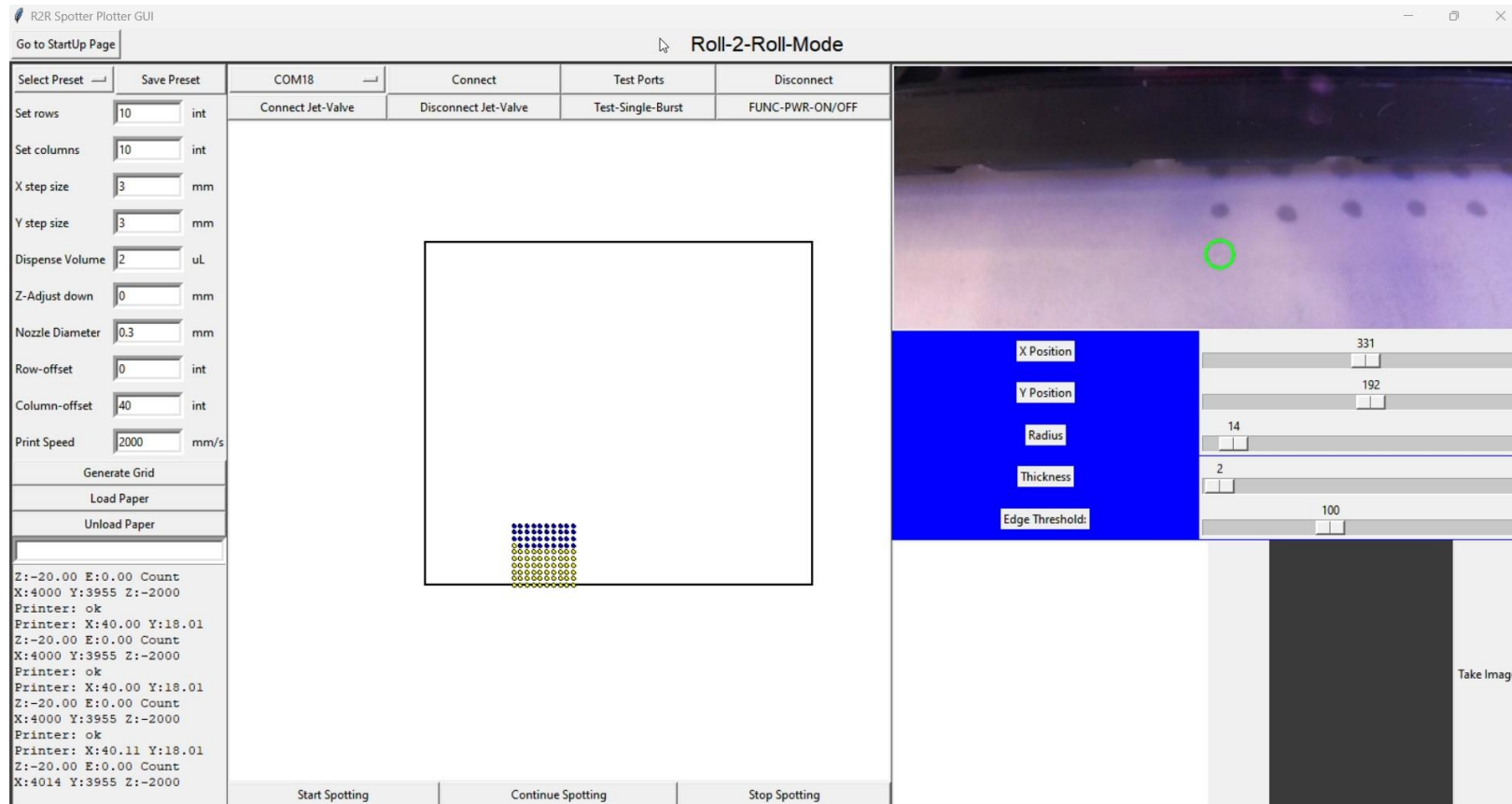
- Non-contact → Limited risk of crashing (no sensitive components)
- Handles uneven substrates
- High frequency spotting
- “infinite substrates” by use of R2R

R2R-Spotter Plotter:

what is possible to achieve with a DIY Jet-Valve?



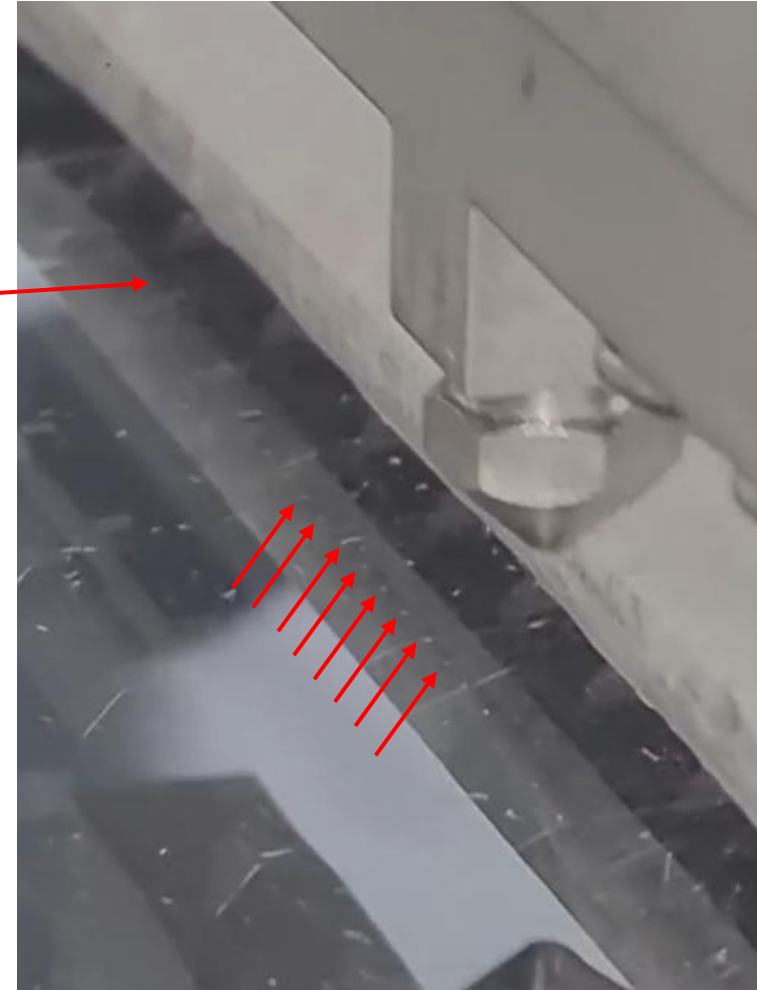
R2R-Spotter Plotter: Mineral oil spotting



Minimum droplet diameter:

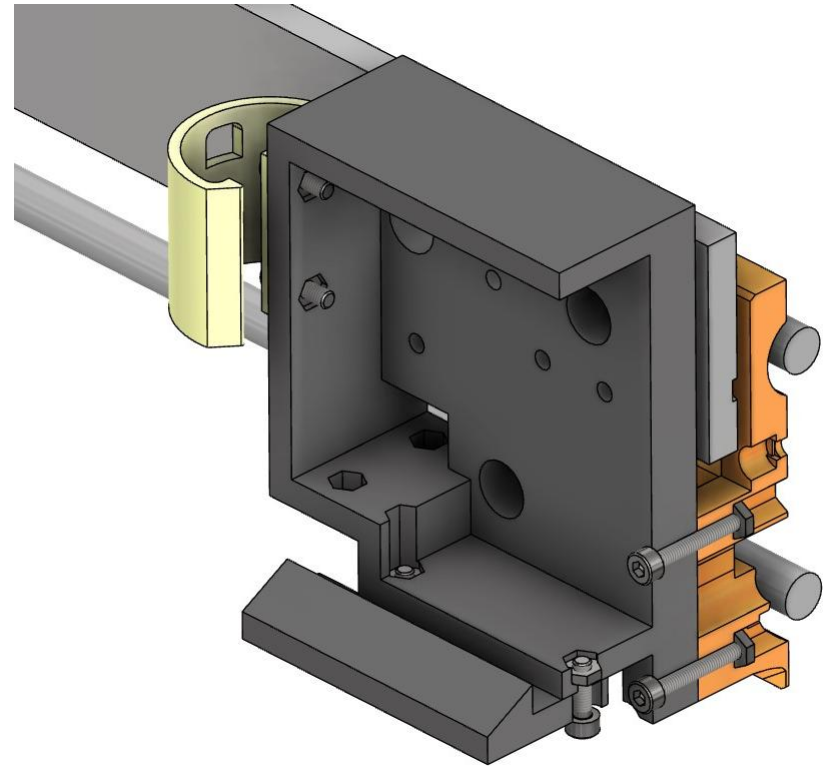
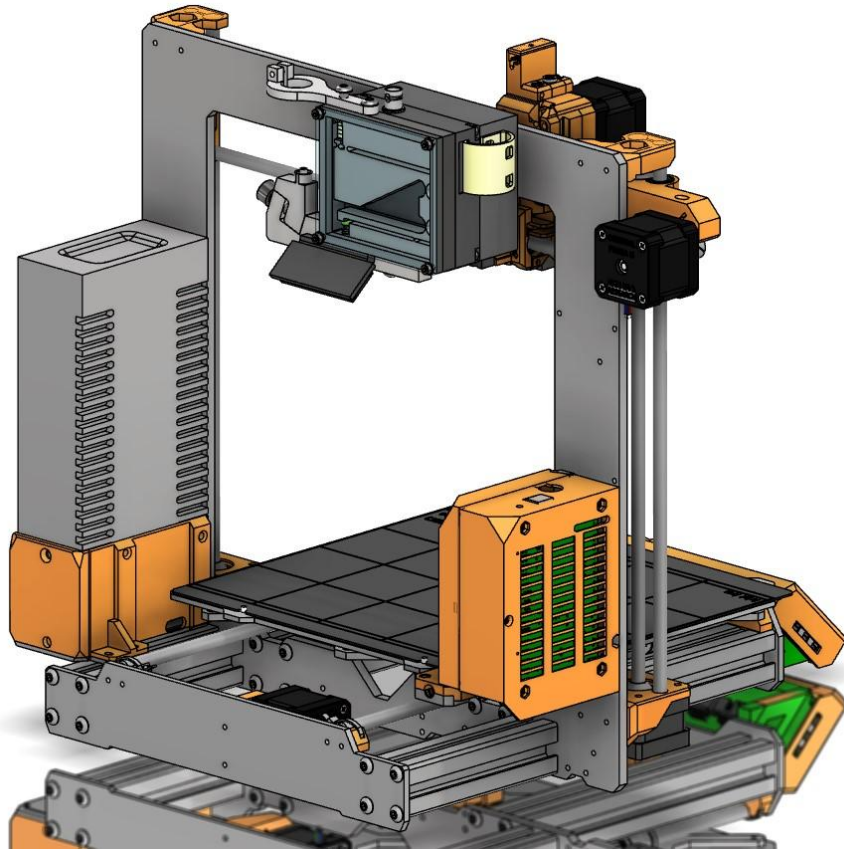
New achievements:

- 301 microns
 - At 15Hz oscillation mode
- 655 microns
 - In Drop-on-Demand mode



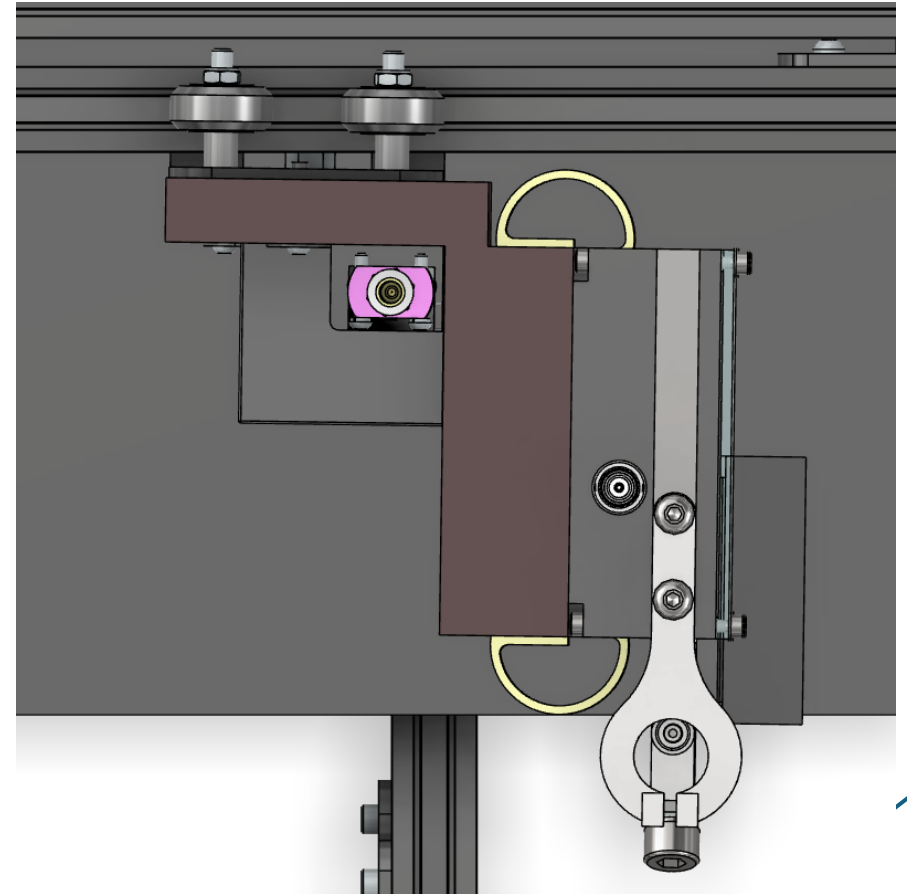
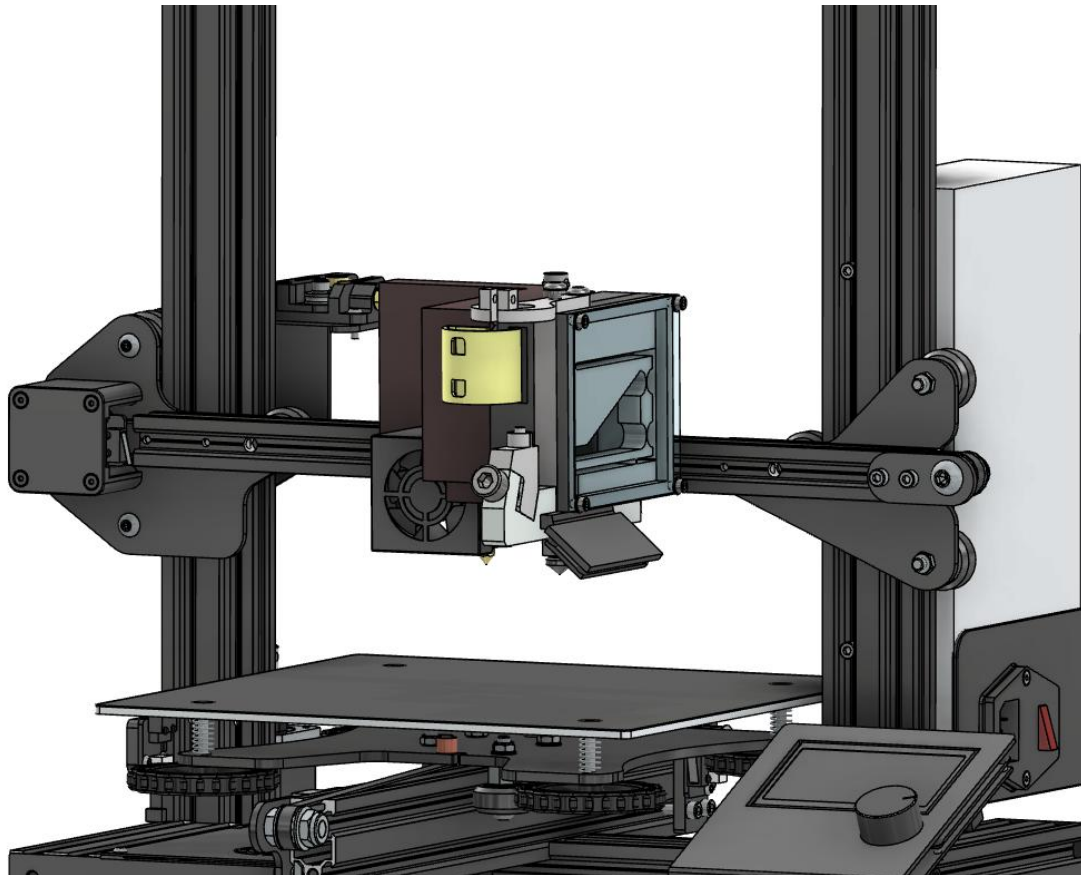
The add-on requirement:

Prusa Mk3 example



The add-on requirement:

Ender 3 example

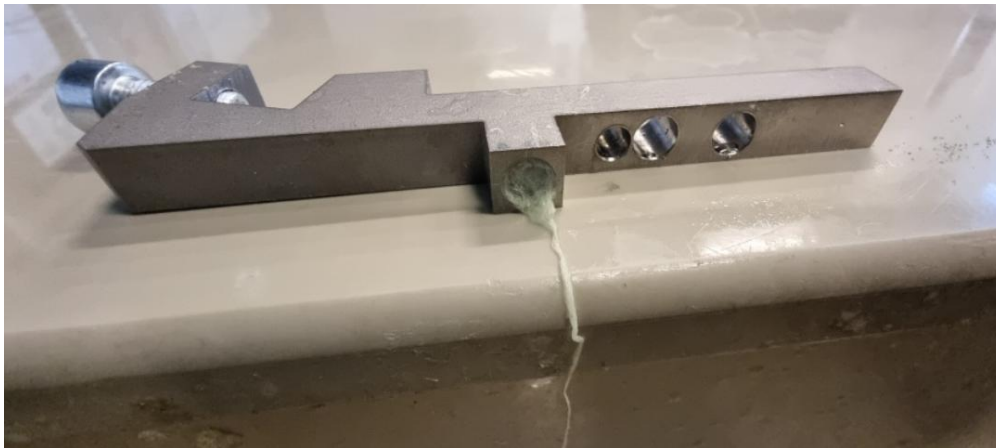


Dry run presentation :

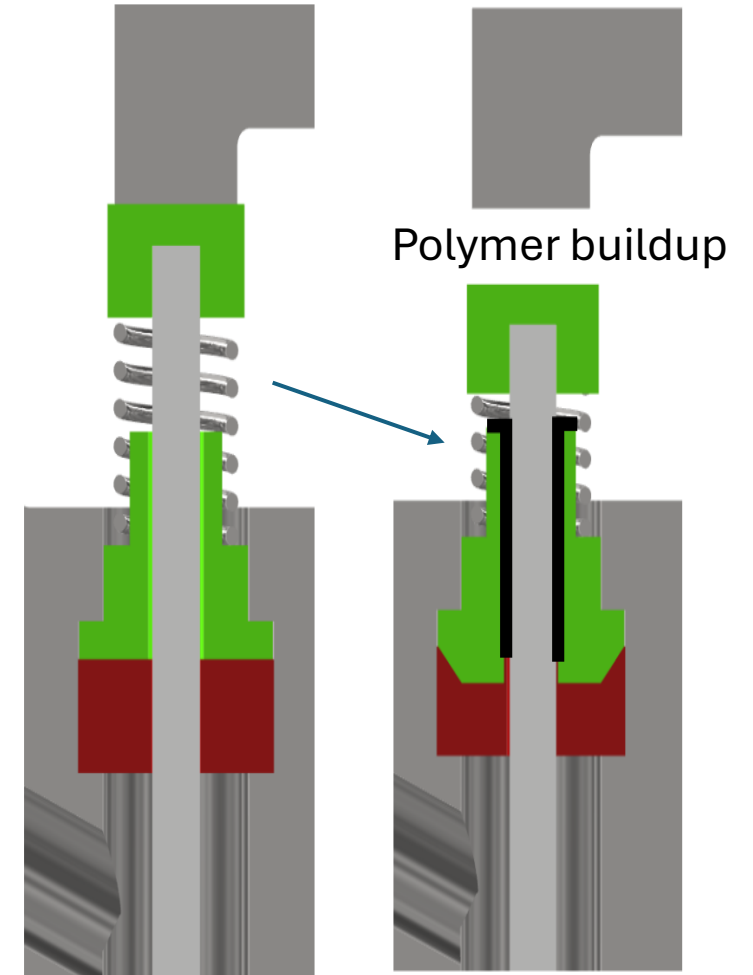
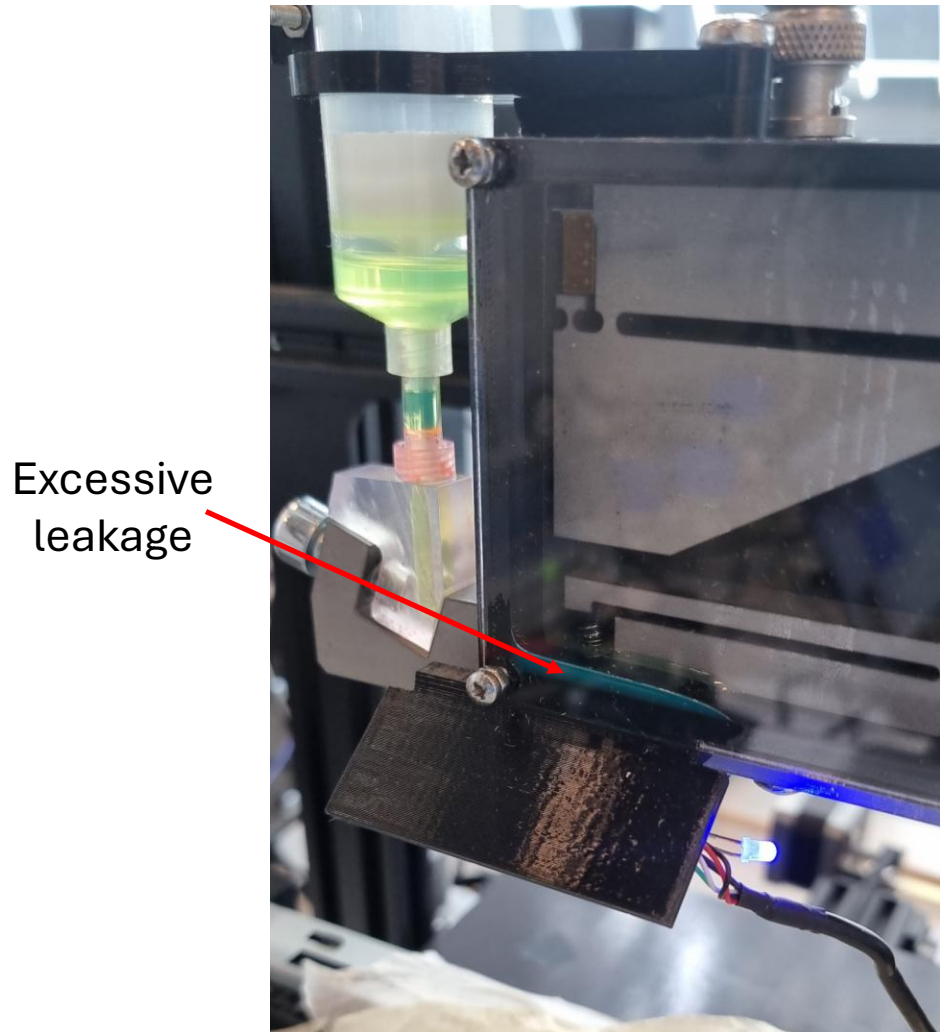


Problems of the system

- Why a syringe pump does not work
 - Relative altitude
 - Stepwise pressure
- Importance of constant pressure
- Only one chemical per Jet reasonable

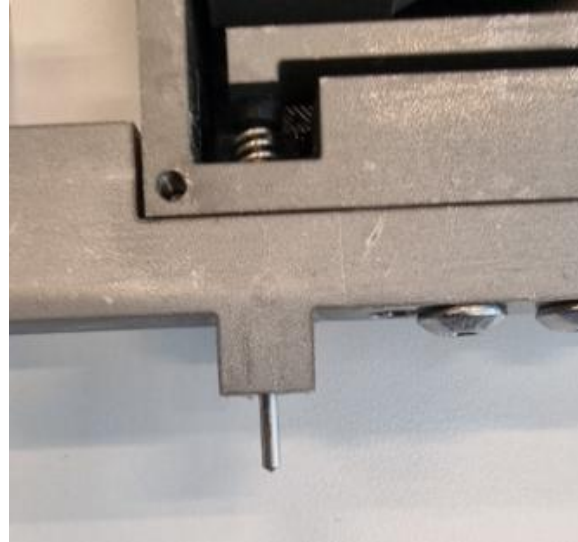
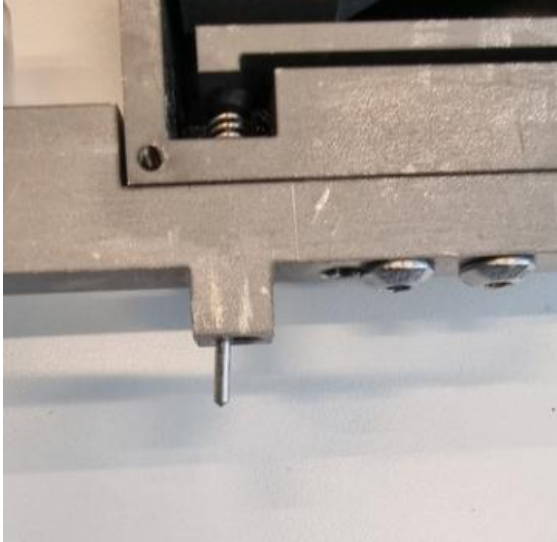


Sealing failure:



How to improve :

Bad mechanical design



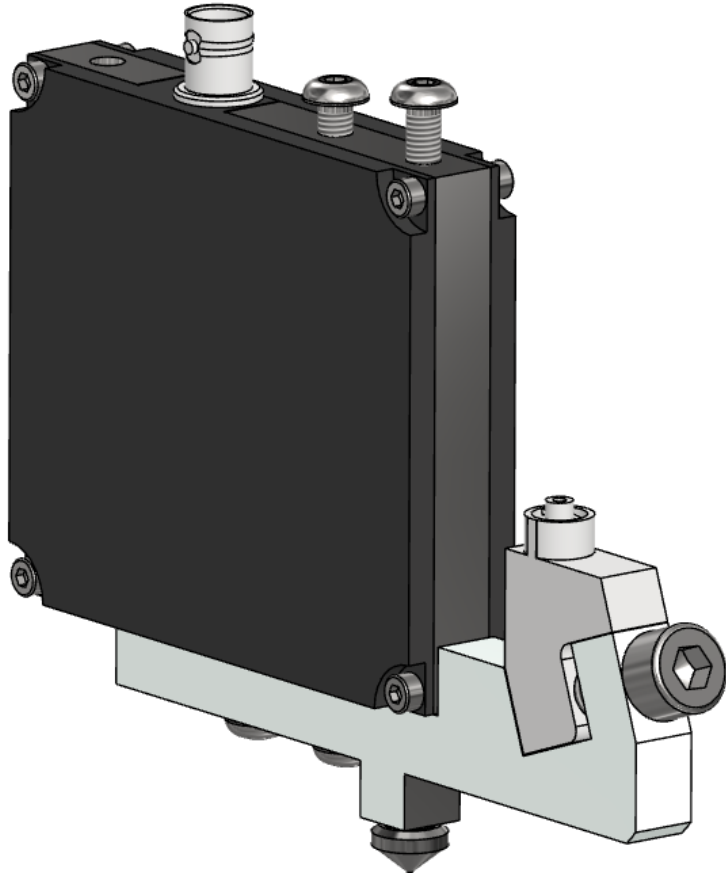
Threaded needle assembly & performance oil sealings



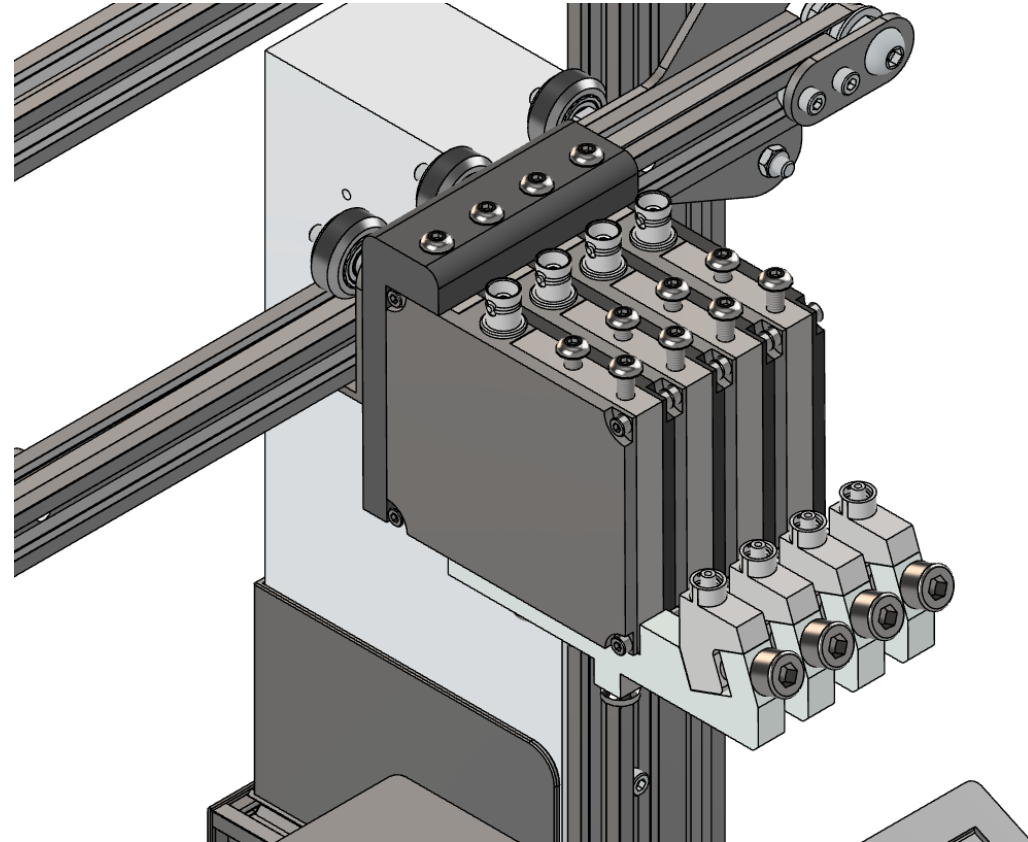
Available for 60\$ per assembly

How to improve : Multiplexing

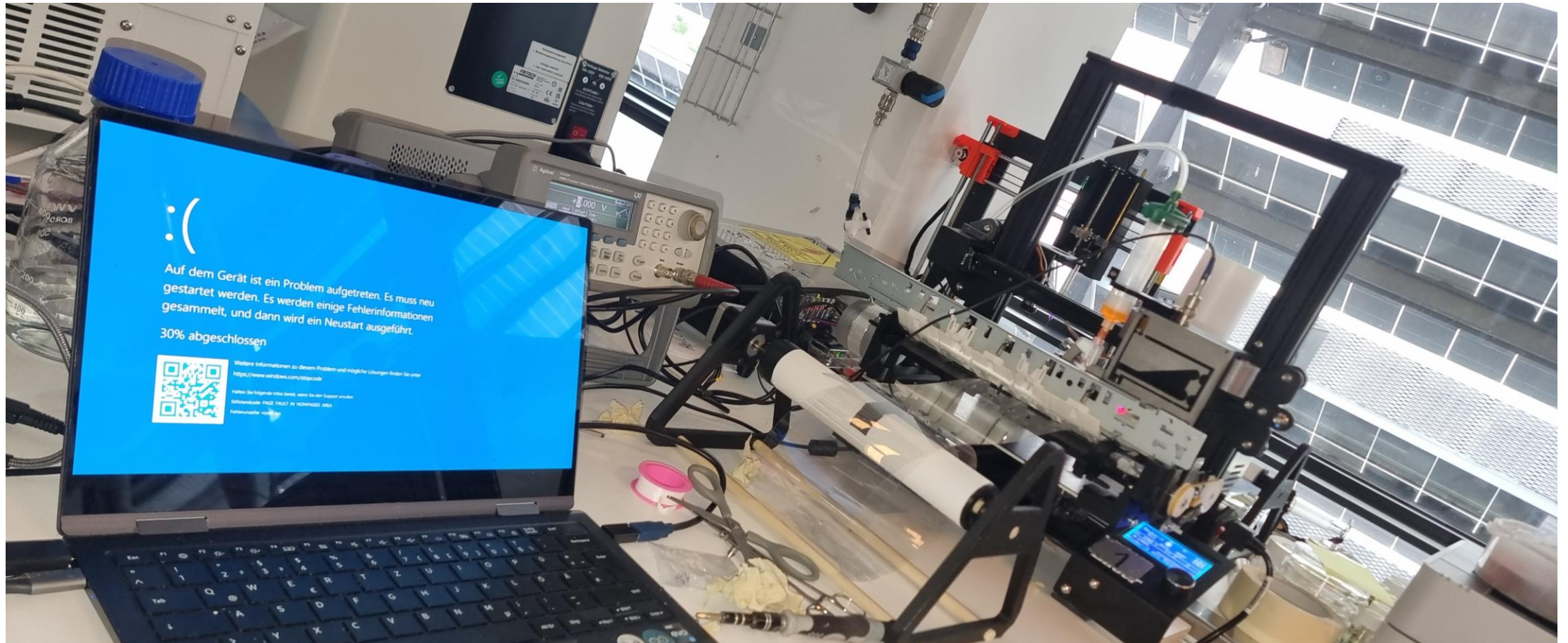
Slim Unit



Parallel Array



Questions:



Budget:

Bill of Materials

Component	Link	Group	Status	Price	Quantity	Total Price	Beneficiary	Amount DKK
Ender 3 Pro	NA	Spotting	Sourced	0	1	0	SDU	1000
SKR CR6 v1	https://	R2R-SKR-CR6	Sourced	250	1	250	Supervisors	6317.585
Nema 17 42 34	https://	R2R	Sourced	0	2	0		7317.585
PowerSupply	https://	R2R	Sourced	143.96	1	143.96		
Vextra 6v Stepper	NA	R2R	Sourced	0	2	0		
USB Webcam	NA	Spotting	Sourced	0	1	0		
							Spend	Budget
							6317.585	0
Bearing 6000-2RS	https://	R2R	Sourced	0	2	0		
Solderlug	https://	Spotting	Sourced	52	1	52		
Piezo-Stack 10um	https://	Spotting-shi	Sourced	477.29	1	477.29		
PSU-Piezo	https://	Spotting-shi	Sourced	5215	1	5215		
Sealing 2mm id	NA	Spotting-shi	Sourced	0	1	0		
BNC-Plug	https://	Spotting	Sourced	23.6	3	70.8		
DIN-Plug	https://	Spotting	Sourced	38.23	1	38.23		
DIN-Wallmount	https://	Spotting	Sourced	33.17	1	33.17		
Wood-Rod	NA	R2R-SKR-CR6	Sourced	20	1	20		
Nozzle	https://	Spotting	Sourced	17.135	1	17.135		

0.3mm droplets, X10 magnification

